Daniel Mulugeta, PhD

Assistant Professor of Physics | Spectroscopy & Nanoscience Researcher

Hawassa University, Ethiopia
Email: danielm@hu.edu.et
Nationality: Ethiopian

Professional Profile

Experimental physicist with international research and teaching experience in Ethiopia, Germany, and the United States. Specialized in nanoscale physics, surface science, laser spectroscopy, and photochemistry on metal nanoparticles. Extensive experience in photoexcitation studies of supported silver nanoparticles using femtosecond and nanosecond laser systems, as well as investigation of ultra-thin metal films on semiconductor surfaces.

Experienced in advanced experimental techniques including STM/S, XPS, LEED, AES, FTIR, UV-Vis spectroscopy, and ultra-high vacuum systems. Current research activities also include FTIR spectroscopy, UV-Vis spectroscopy, and machine learning applications for environmental and materials research. Experienced in interdisciplinary research involving nanomaterials, microplastic pollution analysis, nanoparticle-based sensing, and data-driven scientific analysis.

Core Skills

Scientific & Laboratory Skills

- FTIR Spectroscopy
- UV-Vis Spectroscopy
- Scanning Tunneling Microscopy/Spectroscopy (STM/S)
- X-ray Photoelectron Spectroscopy (XPS)
- Low Energy Electron Diffraction (LEED)
- Auger Electron Spectroscopy (AES)
- Temperature Programmed Desorption (TPD)
- Resonance Enhanced Multiphoton Ionization (REMPI)
- Ultra High Vacuum (UHV) Systems
- Laser Spectroscopy and Surface Photochemistry
- Nanomaterials Characterization

Software & Programming Skills

- Python Programming
- Scientific Data Analysis with Python
- Machine Learning for Spectroscopic Data
- PLS Regression Analysis
- Random Forest Modeling
- Data Visualization
- HTML, CSS, Basic JavaScript
- Microsoft Office Suite
- OriginLab
- LaTeX

Research & Academic Skills

- Experimental Design
- Scientific Writing
- Research Proposal Development
- Academic Presentation
- Student Supervision and Mentorship
- Interdisciplinary Research Collaboration

Education

PhD in Physics (Dr.rer.nat.)

Freie University Berlin, Germany | 2010

Research Director: Prof. Dr. Hans-Joachim Freund
Dissertation: **Size-dependent photochemistry of NO dimers adsorbed on alumina supported silver nanoparticles**

MSc in Physics

University of Stuttgart, Germany | 2005

Research Director: Prof. Dr. Ulrich Starke
Thesis: **Ultra-thin metal films on SiC surfaces**

B.Ed. in Physics

Bahir Dar University, Ethiopia | 2001

Academic and Research Experience

Assistant Professor of Physics

Hawassa University, Ethiopia | March 2020 – Present

- Teaching undergraduate and graduate physics courses
- Supervising student research projects
- Conducting research in spectroscopy, nanoscience, and environmental physics
- Developing machine learning approaches for FTIR spectral analysis
- Leading interdisciplinary research activities

Post-Doctoral Research Associate

University of Tennessee & Oak Ridge National Laboratory (USA) | January 2011 – Present

Research Topic: Electronic properties of sub-monolayer metal films on semiconductor substrates
Research Director: Prof. Hanno H. Weitering

Post-Doctoral Researcher

Fritz-Haber Institute of the Max Planck Society, Germany | July 2010 – December 2010

Research Topic: Photoexcitation of metal nanoparticles on oxide films
Research Director: Prof. Dr. Hans-Joachim Freund

PhD Research Assistant

Fritz-Haber Institute of the Max Planck Society, Germany | March 2006 – July 2010

- Conducted experimental surface science research
- Worked with advanced laser spectroscopy systems
- Performed nanoscale photochemistry experiments

MSc Research Assistant

Max Planck Institute for Solid State Research, Germany | October 2004 – November 2005

- Investigated ultra-thin metal films on SiC surfaces
- Conducted interface analysis experiments

Physics Teacher

Theodros II Senior Secondary School, Ethiopia | September 2001 – June 2003

Research Interests

- Nano-scale Physics
- Spectroscopy
- Surface Science
- Experimental Condensed Matter Physics
- Quantum Optics
- Laser Spectroscopy
- Environmental Spectroscopy
- Machine Learning for Spectroscopy
- Nanomaterials and Sensors
- Green Nanotechnology

Research Outputs and Scientific Contributions

Software and Data Analysis Projects

- Python-based FTIR spectral analysis tools
- Machine learning models for polymer identification
- PLS regression models for fuel adulteration analysis
- Random forest classification for microplastic source identification
- Spectroscopic data preprocessing and visualization workflows

Ongoing Research Projects

- Green synthesis of CuNP-based sensors for food adulteration detection
- FTIR and machine learning analysis of microplastic pollution in irrigated farmlands
- Spectroscopic assessment of sunscreen potential in Ethiopian medicinal plants
- Fuel adulteration detection using spectroscopy and machine learning

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Selected Publications

1. Daniel Mulugeta, Ki Hyun Kim, Kazuo Watanabe, Dietrich Menzel, Hans-Joachim Freund. *Size Effects in Thermal and Photochemistry of (NO)2 on Ag Nanoparticles.*

   Phys. Rev. Lett. 101, 146103 (2008)

2. Daniel Mulugeta, Kazuo Watanabe, Dietrich Menzel, Hans-Joachim Freund. *State-resolved investigation of the photodesorption dynamics of NO from (NO)2 adsorbed on Ag NPs of various sizes in comparison with Ag(111).* **J. Chem. Phys. 134, 164702 (2011)**
3. Ki Hyun Kim, Kazuo Watanabe, Daniel Mulugeta, Hans-Joachim Freund, and Dietrich Menzel. *Enhanced Photoinduced Desorption from Metal Nanoparticles by Photoexcitation of Confined Hot Electrons using Femtosecond Laser Pulses.* **Phys. Rev. Lett. 107, 04740 (2011)**

4. Dietrich Menzel, Ki Hyun kim, Daniel Mulugeta, Kazuo Watanabe. *Case studies in surface photochemistry on metal nanoparticles.* **J. Vac. Sci. Technol. A31(5), Sep/Oct 2013**
5. Daniel Mulugeta, Fangfei Ming, Weisong Tu, Tyler S. Smith, Ying-Tzu Huang, ReneeD. Diehl, Paul C. Snidjers, Hanno H. Weitering. *Hidden phase in a two dimensional Sn layer stabilized by modulation hole doping.* **Nat. Communs 8, 147214 (2017)**
6. Fangfei Ming, Steven Johnston, Daniel Mulugeta, Tyler S. Smith, Paolo Vilmercati,Guenseop Lee, Thomas A. Maier, Paul C. Snidjers, Hanno H. Weitering. *Realization of mott insulating Sn layer on silicon surface induced by modulation hole doping.* **Phys. Rev. Lett. 119, 266802 (2017)**
7. Seho Yi, Fangfei Ming, Ying-Tzu Huang, Tyler S. Smith, Xiyou Peng, Weisong Tu, Daniel Mulugeta, Renee D. Diehl, Paul C. Snider's, Jun-Hyung Cho, and Hanno H.Weltering. *Atomic and electronic structure of doped Si(111) (2√3×2√3)R30o -Sn interfaces.* **Phys. Rev. B 97 1954 (2018)**

Conference Presentations and Invited Talks

- Presented research at major international conferences in Europe and United States
- Presentations delivered at American Physical Society (APS) March meetings, Gordon Research Conferences, European Conference on Surface Science, International Conference on Nanoscience and Technology, and other international scientific meetings
- Invited and contributed talks on nanoparticle photochemistry, thin films, surface science, laser spectroscopy, and nanoscale physics
- More than 25 international conference presentations and scientific contributions
- Full list of conference presentations and invited talks available upon request

Awards and Recognition

- Finalist, Nottingham Prize Competition at the 71st Physical Electronics Conference, University at Albany-SUNY, USA

Languages

- Amharic | English | German

References

Available upon request.